RedCup Eligibility & Shortlist Algorithm

Build spec v1 for Om · from Harsh · 16 October 2026 · launch scope marked where it differs from later versions

1. What this builds

Creators should see, before applying, whether they are eligible for a campaign and exactly why not — like Superset shows students which job posts they can apply to. Eligibility works as four rings, from widest to narrowest:

Ring	What it decides	Who decides	Automated at launch?
1. RedCup verified	Can this creator apply to anything?	System	Yes
2. Campaign must-haves	Can this creator apply to this campaign?	System, using the brand's criteria	Yes
3. Shortlist	Which applicants does the brand see, in what order?	Fit Score + RedCup review	Score yes; quality rating by hand
4. Award	Who gets the campaign?	Brand (48-hour deadline)	Yes, with auto-award fallback

Deliverables:

- A verification gate (Ring 1) with reason codes.
- A per-campaign **criteria** object; each rule is **must** (blocks applying) or **nice** (only affects Fit Score).
- An eligibility engine returning a status + reasons for every creator × open campaign, refreshed daily and on demand.
- Creator-side status on every campaign card, including "almost eligible".
- A live "≈ N creators eligible" count in the brand's campaign builder.
- Fit Score, shortlist generation, brand selection with deadline, and application states.

2. Data inputs

Store the latest creator numbers in a **creator_metrics** table, refreshed daily from the Instagram API (and right after a creator connects or reconnects).

Field	Source	Notes
followers, account_type, media_count	Instagram API	account_type must be Creator or Business
account_age_days	Instagram API (proxy)	Instagram doesn't expose account creation date: use the date of the oldest post returned
avg_reel_views_10	Instagram API	Mean views of the last 10 reels; null if fewer than 3 reels
engagement_rate_10	Instagram API	Mean of (likes + comments + shares + saves) ÷ views over the last 10 reels

Field	Source	Notes
audience_country_pct, audience_city_pct, audience_gender_age	Instagram audience insights	Only available above Meta's minimum follower threshold: store null when unavailable, never 0
niches (max 3), niche_verified[]	Creator profile + RedCup review	Self-declared; a niche counts as verified once RedCup checks recent reels
languages, city	Creator profile	
whatsapp, pan, upi	Creator profile	Required for Ring 1
redcup_score, on_time_rate, first_draft_rate	RedCup data	Null for creators with no completed campaigns
quality_rating (1–5)	RedCup team	Manual at launch; null if not yet rated
fraud_flags_upheld, competitor_history	RedCup data	Competitor history = brands the creator was awarded on RedCup, with dates

3. Ring 1 — RedCup verified

Run on sign-up, on every metrics refresh, and before every application. All rules must pass. Return every failed rule, not just the first, so the creator can fix them all at once.

Rule	Pass condition	Reason code	Message shown
IG_CONNECTED	Professional account connected, token valid	IG_NOT_CONNECTED	Connect your Instagram professional account
ACCOUNT_AGE	account_age_days ≥ 90	ACCOUNT_TOO_NEW	Your account needs to be 90 days old (yours: {n} days)
MIN_POSTS	media_count ≥ 10	TOO_FEW_POSTS	You need at least 10 posts (you have {n})
PROFILE_COMPLETE	WhatsApp, PAN, UPI all present	PROFILE_INCOMPLETE	Add your {missing} to apply
NO_FRAUD	fraud_flags_upheld = 0	ACCOUNT_RESTRICTED	Your account can't apply right now; contact support

4. Campaign criteria

Each campaign stores a **criteria** JSON object. Every rule has a **mode**: **must** or **nice**. Numeric rules support min and/or max.

```
{
  "followers":        { "mode": "must", "min": 10000, "max": 50000 },
  "niches":           { "mode": "must", "any_of": ["beauty", "skincare"], "require_verified": false },
  "avg_reel_views":   { "mode": "nice", "min": 8000 },
  "engagement_rate":  { "mode": "must", "min": 0.02 },
  "audience_country": { "mode": "must", "country": "IN", "min_pct": 60 },
  "audience_cities":  { "mode": "nice", "any_of": ["Delhi", "Mumbai"], "min_pct": 20 },
  "languages":        { "mode": "must", "any_of": ["hi", "en"] },
  "audience_gender":  { "mode": "nice", "gender": "female", "min_pct": 55 },
  "competitor_lock":  { "mode": "must", "brand_ids": ["b_123", "b_456"], "days": 30 },
  "min_redcup_score": { "mode": "nice", "min": 60 }
}
```

Brand builder presets (fill criteria in one click, then editable): "Beauty · micro · pan-India", "Food · nano · Hindi", "Fashion · micro · metro cities". Store presets as JSON templates.

5. Eligibility engine

Evaluate every **must** rule. **nice** rules never block — they only feed the Fit Score. Status precedence (first match wins):

Status	When	Creator sees
NOT_VERIFIED	Any Ring 1 rule fails	Finish your profile: {Ring 1 messages}
INCOMPLETE	A must rule needs data that's null because the creator hasn't connected or completed something they control	Connect Instagram / add details to check eligibility
NOT_ELIGIBLE	Any must rule fails beyond tolerance, or data is unavailable from Instagram	Not eligible: {reason for each failed rule}
ALMOST	Exactly one numeric must rule fails, and the gap is ≤ 10% of the threshold	Almost: "You're 800 followers away"
ELIGIBLE	All must rules pass	Apply

```
function evaluate(creator, campaign):
    r1 = ring1_failures(creator)
    if r1: return NOT_VERIFIED, r1

    fails = []; missing = []
    for rule in campaign.criteria where rule.mode == "must":
        value = metric(creator, rule)
        if value is null:
            if creator_can_fix(rule): missing.append(rule)          # IG not connected
            else: fails.append(reason(rule, "DATA_UNAVAILABLE"))     # no audience data
            continue
        if not passes(rule, value):
            fails.append(reason(rule, value))

    if missing:   return INCOMPLETE, missing
    if not fails: return ELIGIBLE, []
    if len(fails) == 1 and fails[0].numeric and gap_pct(fails[0]) <= 0.10:
        return ALMOST, fails
    return NOT_ELIGIBLE, fails

gap_pct(rule) = |threshold - value| / threshold        # 9,200 vs min 10,000 -> 0.08 -> ALMOST
```

Reason messages must name the rule, the requirement and the creator's value: "Needs 10K–50K followers · you have 6.2K", "Needs 60%+ audience in India · yours is 48%", "Needs a Beauty or

Skincare niche", "You promoted a competing brand in the last 30 days", "Instagram doesn't share audience data for accounts your size yet".

6. When to compute, and locking

- **Daily job:** after the metrics refresh, recompute eligibility for every verified creator × every open campaign. Store in **eligibility_results**.
- **On demand:** recompute for one creator when they connect Instagram, edit their profile, or open a campaign card older than 24 hours.
- **New campaign:** compute for all creators when a campaign goes live, then send WhatsApp alerts to ELIGIBLE creators only (and optionally ALMOST, as "you're close").
- **Lock on apply:** when a creator applies, copy the criteria and their metrics into the application (**criteria_snapshot**, **metrics_snapshot**). Later drops in followers don't remove them.
- **Feed filter:** creators can toggle "Eligible only" / "Show all"; default is Eligible + Almost on top.
- **Brand live count:** as the brand edits criteria, run a count over **creator_metrics** with the must rules (debounce ~500 ms). Show "≈ N creators eligible". If N < 3 × slots, warn: "Few creators match — try relaxing a must-have".

7. Fit Score (Ring 3)

Computed for every application when the application window closes (or when the brand asks for the shortlist early). Each part is 0–1; Fit Score = 100 × weighted sum.

Part	Weight	Calculation (v1)
A. Niche & audience match	40%	0.5 × niche (primary match 1.0, secondary 0.7, verified +0.1 capped at 1) + 0.3 × min(target-country %, 100) ÷ 100 + 0.2 × share of **nice** rules met
B. Past performance	30%	If redcup_score exists: redcup_score ÷ 100. Else: percentile rank, among this campaign's applicants, of avg_reel_views_10 ÷ followers
C. Content quality	20%	(quality_rating − 1) ÷ 4; 0.5 if not yet rated
D. Reliability	10%	0.6 × on_time_rate + 0.4 × first_draft_rate; 0.7 for creators with no history (new creators aren't punished)

```
fit = 100 * (0.40*A + 0.30*B + 0.20*C + 0.10*D)
tie-break: higher avg_reel_views_10, then earlier applied_at
store fit_breakdown = {A, B, C, D} on the application (brands and admins see the parts)
```

Keep the weights in config, not code — we'll tune them after the first 20–30 campaigns.

8. Shortlist and award (Rings 3–4)

- **Shortlist size:** top min(applicants, 3 × slots) by Fit Score (configurable 2–3×).
- **Admin review (launch):** RedCup can pin or remove applicants before sending, with a required reason. Log every override.
- **Brand view:** each shortlisted creator's profile, last 6 reels, followers, avg views, engagement, audience split, Fit Score and its parts.
- **Award deadline:** brand selects up to **slots** creators within 48 hours. If not, auto-award the top Fit Scores and notify the brand.
- **On award:** reserve the creator's cap from the budget (ledger entry), send direction + brief by WhatsApp, start the draft deadline.
- **Waitlist:** if an awarded creator declines or misses the draft deadline, release the reservation and offer the slot to the next shortlisted creator.
- **Not selected:** notify with a friendly message; never show their Fit Score to them in v1.

Application states

State	Moves to	Trigger
APPLIED	SHORTLISTED / NOT_SHORTLISTED	Shortlist generated
SHORTLISTED	AWARDED / WAITLISTED / NOT_SELECTED	Brand selection or 48-hour auto-award
WAITLISTED	AWARDED / NOT_SELECTED	A slot frees up, or the campaign fills
AWARDED	DRAFT_SENT → (existing claim flow: approved, posted, tracking, paid)	Creator accepts and uploads a draft
AWARDED	EXPIRED	Draft deadline missed — slot goes to the waitlist

9. Tables to add

```
creator_metrics(creator_id, followers, account_type, account_age_days, media_count,
                avg_reel_views_10, engagement_rate_10, audience_country_pct jsonb,
                audience_city_pct jsonb, audience_gender_age jsonb, refreshed_at)
campaigns      (+ criteria jsonb, slots, application_closes_at, shortlist_multiplier, preset_id)
eligibility_results(campaign_id, creator_id, status, reasons jsonb, computed_at)   -- unique (campaign_id, creator
applications   (campaign_id, creator_id, status, applied_at, criteria_snapshot jsonb,
                metrics_snapshot jsonb, fit_score, fit_breakdown jsonb, award_deadline_at)
shortlist_overrides(application_id, action pin|remove, reason, admin_id, created_at)
criteria_presets(id, name, criteria jsonb)
```

10. Test cases

#	Creator	Campaign must-haves	Expected
1	8,000 followers, Beauty, IG connected	Followers 10K–50K, Beauty	NOT_ELIGIBLE: 20% gap > 10%
2	9,200 followers, Beauty	Followers 10K–50K, Beauty	ALMOST: "800 followers away"
3	25,000 followers, Food	Followers 10K–50K, Beauty	NOT_ELIGIBLE: niche
4	Instagram not connected	Any	NOT_VERIFIED: connect Instagram
5	3,000 followers, no audience data from Instagram	Audience 60%+ India (must)	NOT_ELIGIBLE: data unavailable
6	30,000 followers, awarded by a competitor 12 days ago	Competitor lock 30 days	NOT_ELIGIBLE: competitor
7	Eligible, applies, then drops to 9,500 followers	Followers 10K–50K	Stays APPLIED (locked snapshot)
8	Two failed numeric rules, both within 10%	Followers + engagement	NOT_ELIGIBLE (ALMOST needs exactly one)
9	Brand doesn't select within 48 hours	Slots = 5	Top 5 Fit Scores auto-awarded

11. Launch vs later

At launch	Later
Rings 1–2 fully automated, status + reasons on every card, ALMOST state	Niche detection from captions and audio, not just self-declared
Live eligible count and presets in the brand builder	Fit Score weights tuned on real campaign data
Fit Score with manual quality rating; admin pin/remove	Automatic quality rating; RedCup Score as a must-have option
48-hour award deadline with auto-award and waitlist	Lookalike recommendations: "creators like your top performers"

Questions or changes: discuss with Harsh before building anything that changes what creators or brands see. Thresholds (90 days, 10 posts, 10% tolerance, 3× shortlist, 48 hours) are starting values — keep them in config.